Helicopter Emergency Rescue

GDP 2025

4.1 Team Organization

Although smaller sub-groups or individual students may be responsible for only one disciplinary aspect of the design, the entire group's success in this multidisciplinary design project relies on effective communication and collaboration of all group members.

In recognition of this, each group will typically have some students assigned or elected to fill a project management role. These students will act as coordinators to facilitate the running of the project. They will also act as mediators to solve design conflicts that may arise between individual member's design choices, but if an external decision is needed, it will need to be taken by the whole team through an appropriate voting mechanism.

To ensure that all students are kept aware of the group's progress, and how that might have an impact on their areas of responsibility, it is recommended that a short update meeting is held daily, where progress can be presented, design decisions discussed and new goals can be set.

4.2 Weekly Progress Meetings

Each team is expected to hold a weekly progress meeting where all group members and supervising academics will be in attendance. These meetings will be called by the project managers, who will organize the agenda according to the issues that need discussion. A schedule should be available to the academic advisors at the start. Times and dates are to be set in discussion with the supervising academics.

These meetings will offer both students and academics the opportunity to review the past week's progress and discuss any significant problems with their colleagues and staff, in order to reach generally acceptable decisions.

Students will prepare presentations for the weekly group meeting and will be given 10 minutes to present, followed by a 5-minute Q&A. Note that, not all students need to present each week, however each group member must have presented at least twice by the end of the project.

The content of these presentations will in part inform each student's Carrying Out mark, but will also offer an opportunity for students to receive feedback on their work and presentation skills. In addition to the feedback received during these presentations, students should aim to meet with their individual supervisors once a week, following the progress meetings or at a mutually convenient time.

5.1 Project Outcomes / Deliverables

Students are expected to deliver the following pieces of work by the end of the project to report the project outcome and their contributions towards it.

- **Design specification document:** A single collaborative document which details the overall results of the project and the contributions of each subgroup towards achieving them. It can be considered as a summary of the entire project to which each member contributes and should be no longer than 5 pages in total.
- **Poster:** An A1 landscape poster or equivalent video, presenting a top level view of the projects result, detailing in an approachable way the most important design features and characteristics.
- **Individual report:** A technical report by each member of the group detailing their contribution to the project.
- **Group presentation:** A group technical presentation, presenting, detailing and discussing the resulting design.

Submission Information for Project Reports and Posters

Final Report – Monday 16th June 2025

- *ONE* pdf file of your individual report to be submitted via your report submission box on Blackboard.

Executive Summary – Monday 16th June 2025

- *ONE* pdf file of the group's design specification document to be submitted via the Executive Summary submission box on Blackboard (no hard copies to be submitted with your reports).

Group Poster – Thursday 19th June 2025

- *ONE* pdf of an A1-size poster to be submitted via the Poster submission box on Blackboard. You may wish to print a copy for your group presentation and/or pitch.

	Last Name	First Name	Role	Supervisor
Coordinators	Jain	Tarang	HER01	Ribera Vicent, Maria Dr
	Li	Cheong Sean	HER01	Ribera Vicent, Maria Dr
Aerodynamics	Chen	Yinuo Nick	HER02	Ribera Vicent, Maria Dr
	Cooper	Daniel	HER02	Steiros, Kostas Dr
	Liu	Zhaotong Jim	HER02	Steiros, Kostas Dr
	Shao	Wenyue Summer	HER02	Steiros, Kostas Dr
	Yeo	Chiaw	HER02	Steiros, Kostas Dr
Structural Design & CAD	Bollen Gandolfo	Clara	HER03	Shamsuddin, Siti Ros Dr
	Chabanel	Tristan	HER03	Shamsuddin, Siti Ros Dr
	Pratt	Mercury	HER03	Shamsuddin, Siti Ros Dr
	Xue	Wanning	HER03	Seifikar, Masoud Dr
Systems & Propulsion	Hannan	Sayeed	HER04	Fasel, Urban Dr
	Hassan	Sabina	HER04	Fasel, Urban Dr
	Robson	Ash	HER04	Seifikar, Masoud Dr
	Tan	Tze Dylan	HER04	Seifikar, Masoud Dr
Flight Mechanics, Performance & Control	Chan	Chun Johnny	HER05	Armanini, Sophie Dr
	Harvey	Alex	HER05	Armanini, Sophie Dr
	Li	Hanchen	HER05	Armanini, Sophie Dr
	Song	Haoran	HER05	Armanini, Sophie Dr

Roles

FATT01 – Project coordinators
Academic advisor: Dr Ribera Vicent
• Project coordination & information management • Cost estimation and market analysis • Conceptual trade off studies & design • Compliance with all relevant design standards and regulations

FATT02 – Aerodynamics
Academic advisor: Dr Steiros
• Rotor(s) design • Design of lifting surfaces • Airfoil selection/design/optimizations • Fuselage and empennage aerodynamic design. • If present, tail rotor design.

Roles

FATT03 – Structural Design & CAD
Academic advisor: Dr Shamsuddin
• Prediction of design loading conditions • Rotor structural design and dynamic blade behaviour • Selection of aircraft structural layout and sizing of major components • Material and manufacturing method selection • Landing gear design • Hub design • CAD modeling
FATT04 –Systems & Propulsion
Academic advisor: Dr Fasel, Dr Seifikar
• Powerplant type selection & integration • Powerplant sizing and performance prediction • Transmission considerations • Vehicle sub-systems selection/sizing, layout and integration • Sensors and avionics • Weight & Balance Prediction • Medical and mission specific equipment

Roles

FATT05 – Flight Mechanics, Performance & Control
Academic advisor: Dr Armanini, Dr Ribera Vicent
• Mission definition and operations • Vehicle static and dynamic stability prediction • Response to control inputs • Design of flight control system • Aircraft performance prediction


IDEAL PLANES
OR WHAT CAN HAPPEN IF ONE OF THE TEAM GETS ALL THEIR OWN WAY!
FUSELAGE GROUP
WEIGHTS GROUP
MECHANICAL CONTROLS GROUP
COMPUTER AIDED DESIGN GROUP
HYDRAULICS GROUP
PRODUCTION ENGINEERING GROUP
MAINTENANCE GROUP
EQUIPMENT GROUP
ARMAMENT GROUP
WING GROUP
AVIONICS GROUP
AERODYNAMICS GROUP
EMPENNAGE GROUP
POWERPLANT GROUP
STRESS GROUP

Resources

- https://vfs.umd.edu/designGrad.html
- J. G. Leishman, “Principles of Helicopter Aerodynamics”
- W. Johnson, “Rotorcraft Aeromechanics”
- S. Newman, “Foundations of Helicopter Flight”
- S. Newman, “Basic Helicopter Aerodynamics”
- A. Filippone, “Flight Performance of Fixed and Rotary Wing Aircraft”
- G. Padfield, “Helicopter Flight Dynamics”
- A. Bramwell, “Bramwell’s Helicopter Dynamics”